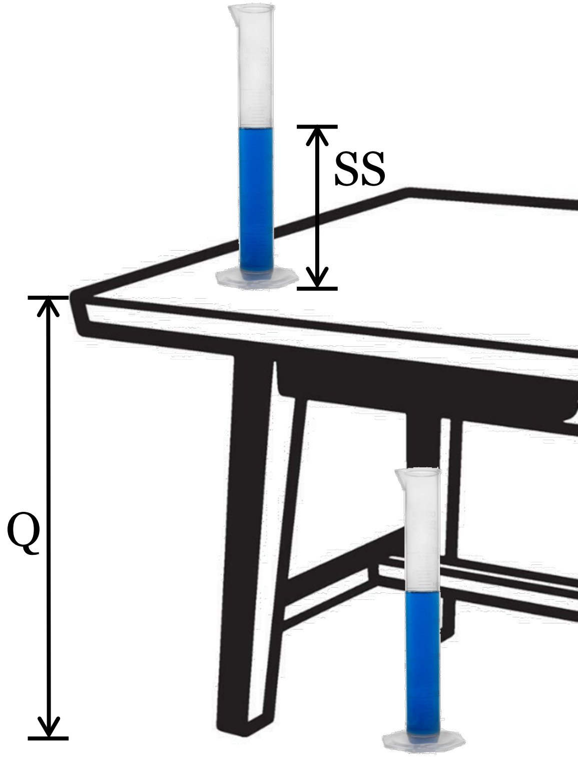
1.3 – DC Operating Point and Signal Variation

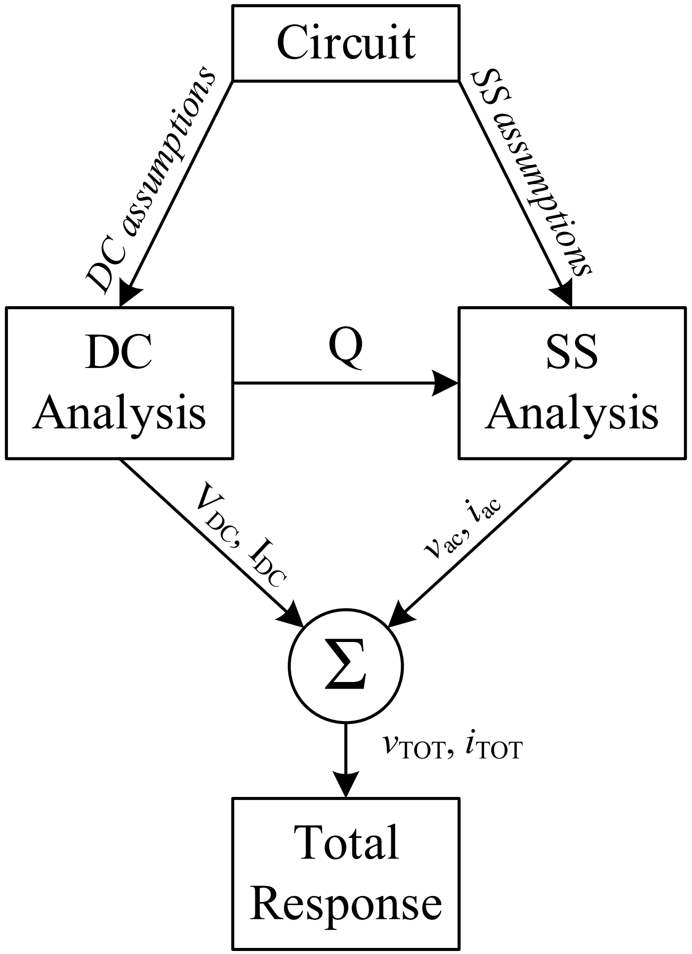
Analyzing Real Circuits

- Real circuits have:
  - A DC Operating Point (Q)
  - A Small-Signal (SS) response (linear only)
  - Often neglect nonlinear AC effects
- Why?
  - To keep signals within the voltage rails
    - Example: A physics experiment (SS) on top of a table (Q) instead of the floor
  - Sometimes the circuit response to SS depends on Q
    - Example: Turning on a gasoline engine (Q) then revving it (SS)



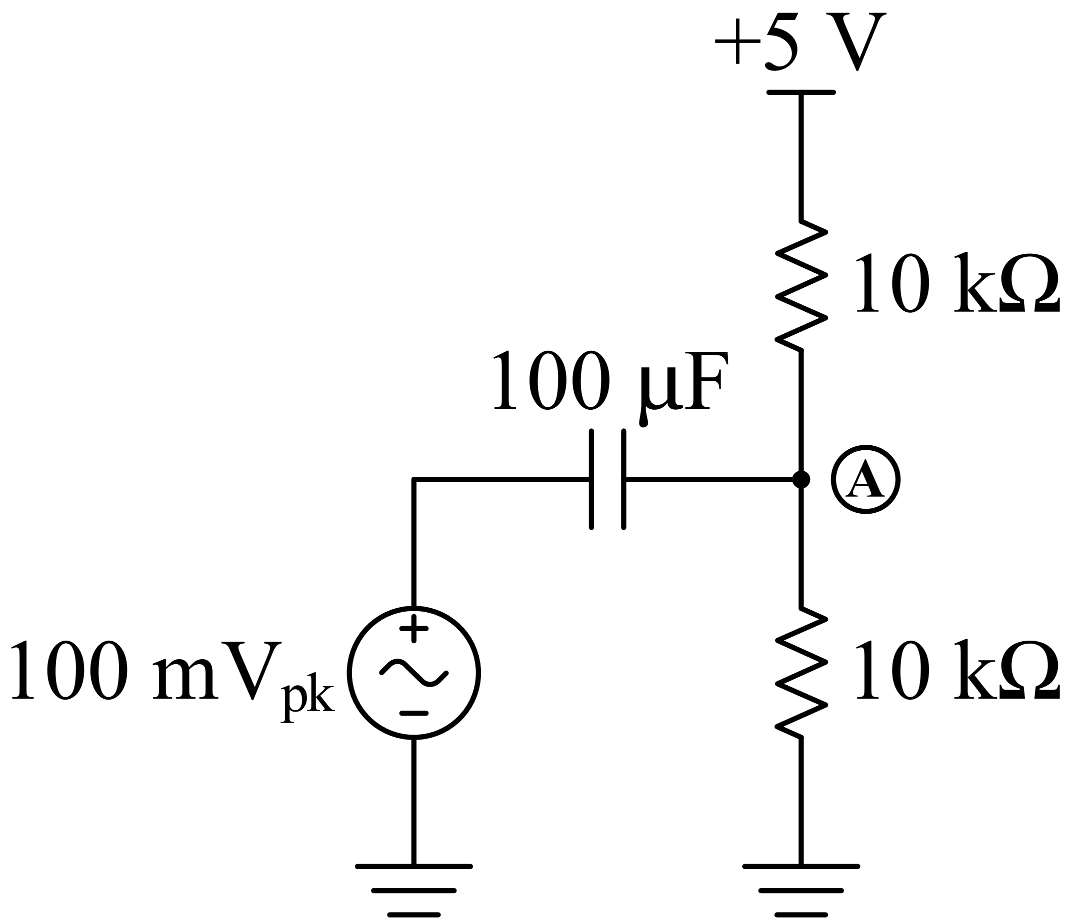
Decomposing Circuit Response

- Draw a DC version of the circuit
  - Determine Q
- Draw a SS version of the circuit
  - Determine SS response
- Sum the responses to get the total response

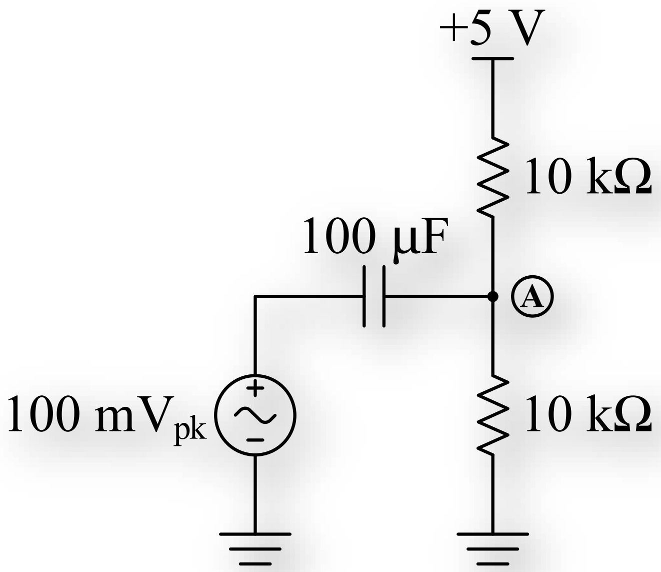


Example

- Determine v_A, the total response of this circuit

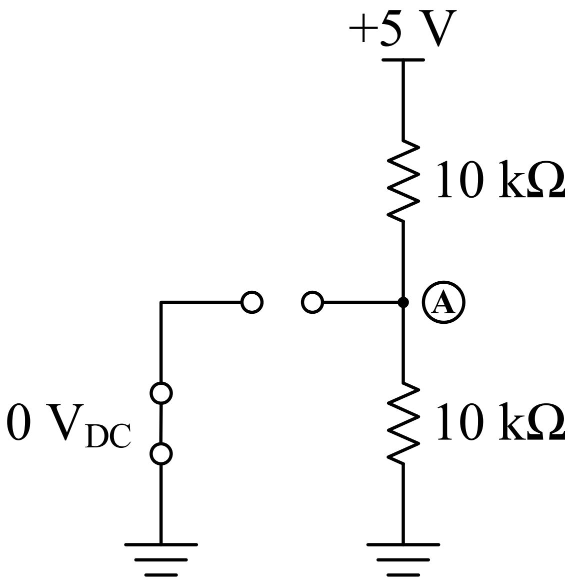


Redraw Circuit for DC and SS Analysis



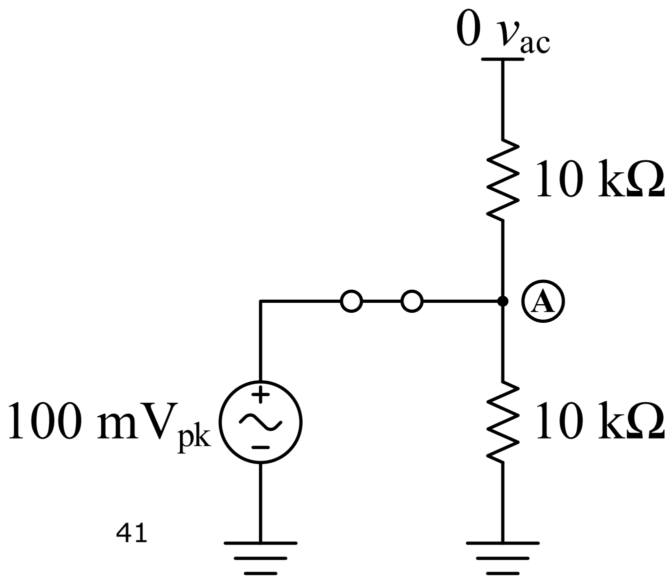
DC:

- C → open
- L → short
- v_{ac} → 0 V_{DC} → short
- i_{ac} → 0 I_{DC} → open

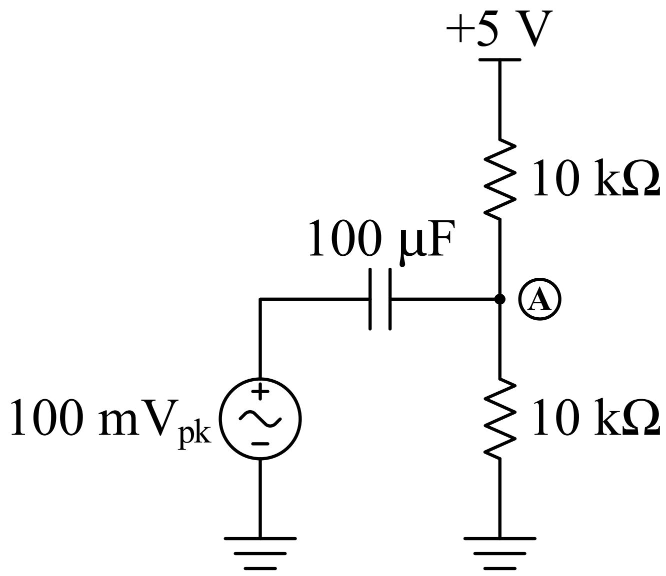


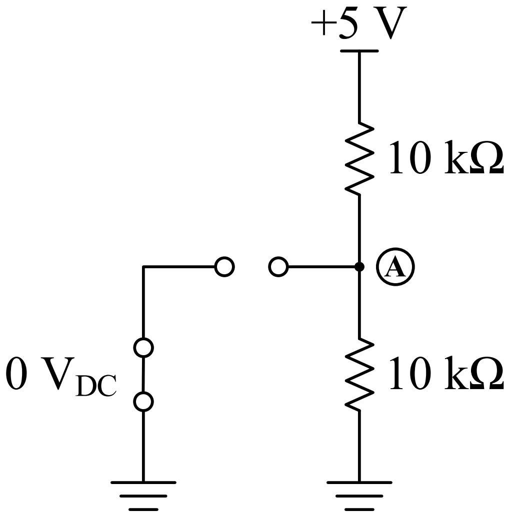
SS (AC):

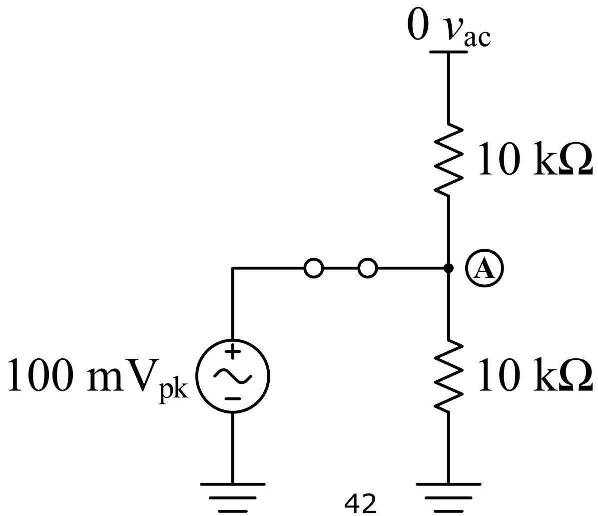
- C → short
- L → open
- V_{DC} → 0 v_{ac} → short
- I_{DC} → 0 i_{ac} → open



Sum to get Total Response







LTSpice Modeling Challenge

- Create the example circuit schematic
  - Perform a DC analysis ("DC op pnt")
  - Perform a transient analysis (choose a frequency for your 100 mV source)
- How could you find only the AC response?
- What does the AC Analysis do, and how is that different?
- What happens if you change the value of C?